Titanic EDA & ML Report

Q1. What is EDA and why is it important?

EDA (**Exploratory Data Analysis**) is the process of exploring and analyzing datasets to understand their structure, patterns, relationships, and anomalies before modeling.

Importance: - Detects missing values, outliers, and errors
- Helps in selecting features
- Guides preprocessing and modeling strategy

Q2. Which plots do you use to check correlation?

- **Heatmap** (shows correlation matrix visually)
- **Pairplot** (visualizes pairwise relationships)
- **Scatter plot** (for two continuous variables)
- **Bar plot** (for categorical vs. numeric relationships)

Q3. How do you handle skewed data?

- **Numerical Data:** Use **median** or transformation (log, sqrt) to reduce skewness.
- **Categorical Data:** Use **mode** to fill missing values.
  In Titanic dataset, `Age` was filled with median and `Embarked` with mode.

Q4. How to detect multicollinearity?

- **Correlation Matrix** → If correlation > 0.8 between features, possible multicollinearity
- **Variance Inflation Factor (VIF)** → VIF > 5 indicates multicollinearity

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Q5. What are univariate, bivariate, and multivariate analyses?

- **Univariate:** Analysis of a single variable (e.g., histogram of `Age`)
- **Bivariate:** Analysis of two variables (e.g., survival rate by `Sex`)
- **Multivariate:** Analysis of more than two variables (e.g., `Age`, `Sex`, and `Pclass` together)

Q6. Difference between heatmap and pairplot?

- **Heatmap:** Matrix-style plot showing correlations using colors
- **Pairplot:** Multiple scatter plots showing pairwise relationships between variables

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Q7. How do you summarize your insights?

- Highlight **key patterns** found in EDA (e.g., women and children had higher survival rates)
- Mention **data quality issues** handled (missing values, duplicates, outliers)
- List **important features** affecting the target
- Provide **model performance results** (accuracy, precision, recall)